

```
(base) alirezahamoudzadeh@Alirezas-MacBook-Air Demo % export OPENROUTER_API_KEY=sk-or-v1-
export AWS_ACCESS_KEY_ID=QA6
export AWS_SECRET_ACCESS_KEY=3AXS8

cd /Users/alirezahamoudzadeh/Downloads/fire/Demo
python demo_client.py
[model called tool] analyze_burn_severity_tool({'min_lon': 14.425, 'min_lat': 40.792, 'max_lon': 14.51, 'max_lat': 40.85, 'pre_start': '2025-05-09', 'pre_end': '2025-08-06', 'post_start': '2025-08-19', 'post_end': '2025-11-16', 'priority_min_area_ha': 0.5})
/opt/anaconda3/lib/python3.12/site-packages/rasterio/warp.py:387: NotGeoreferencedWarning: Dataset has no geotransform, gcps, or rpcs. The identity matrix will be returned.
  dest = _reproject(
Warning 1: HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/05/26/S2C_MSIL2A_20250526T095051_N0511_R079_T33TVF_20250526T155513.S
AFE/GRANULE/L2A_T33TVF_A003767_20250526T095047/IMG_DATA/R10m/T33TVF_20250526T095051_B02_10m.jp2. Retrying again in 3.0 secs
Warning 1: HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/05/26/S2C_MSIL2A_20250526T095051_N0511_R079_T33TVF_20250526T155513.S
AFE/GRANULE/L2A_T33TVF_A003767_20250526T095047/IMG_DATA/R10m/T33TVF_20250526T095051_B02_10m.jp2. Retrying again in 6.0 secs
[09/24/26 15:50:35] WARNING CPLE_AppDefined:HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/06/25/S2C_MSIL2A_20250625T095051_N0511_R079_T33TVF_20250625T152820_SAFE/GRANULE/L2A_T33TVF_A004196_20250625T095707/IMG_DATA/R20m/T33TVF_20250625T095051_B12_20m.jp2. Retrying again in 0.5 secs warp.py:387
WARNING CPLE_AppDefined:HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/06/25/S2C_MSIL2A_20250625T095051_N0511_R079_T33TVF_20250625T152820_SAFE/GRANULE/L2A_T33TVF_A004196_20250625T095707/IMG_DATA/R20m/T33TVF_20250625T095051_B12_20m.jp2. Retrying again in 1.2 secs warp.py:387
[09/24/26 15:50:37] WARNING CPLE_AppDefined:HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/06/25/S2C_MSIL2A_20250625T095051_N0511_R079_T33TVF_20250625T152820_SAFE/GRANULE/L2A_T33TVF_A004196_20250625T095707/IMG_DATA/R20m/T33TVF_20250625T095051_B12_20m.jp2. Retrying again in 2.7 secs warp.py:387
[09/24/26 15:50:39] WARNING CPLE_AppDefined:HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/06/25/S2C_MSIL2A_20250625T095051_N0511_R079_T33TVF_20250625T152820_SAFE/GRANULE/L2A_T33TVF_A004196_20250625T095707/IMG_DATA/R20m/T33TVF_20250625T095051_B12_20m.jp2. Retrying again in 6.0 secs warp.py:387
[09/24/26 15:50:46] WARNING CPLE_AppDefined:HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/06/25/S2C_MSIL2A_20250625T095051_N0511_R079_T33TVF_20250625T152820_SAFE/GRANULE/L2A_T33TVF_A004196_20250625T095707/IMG_DATA/R20m/T33TVF_20250625T095051_B12_20m.jp2. Retrying again in 12.7 secs warp.py:387
[09/24/26 15:50:58] WARNING CPLE_AppDefined:HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/06/25/S2C_MSIL2A_20250625T095051_N0511_R079_T33TVF_20250625T152820_SAFE/GRANULE/L2A_T33TVF_A004196_20250625T095707/IMG_DATA/R20m/T33TVF_20250625T095051_B12_20m.jp2. Retrying again in 25.6 secs warp.py:387
Warning 1: HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/08/19/S2B_MSIL2A_20250819T095029_N0511_R079_T33TVF_20250819T120823.S
AFE/GRANULE/L2A_T33TVF_A044149_20250819T095030/IMG_DATA/R10m/T33TVF_20250819T095029_B03_10m.jp2. Retrying again in 3.0 secs
Warning 1: HTTP error code: 429 - https://eodata.dataspace.copernicus.eu/eodata/Sentinel-2/MSI/L2A/2025/08/19/S2B_MSIL2A_20250819T095029_N0511_R079_T33TVF_20250819T120823.S
AFE/GRANULE/L2A_T33TVF_A044149_20250819T095030/IMG_DATA/R10m/T33TVF_20250819T095029_B03_10m.jp2. Retrying again in 7.0 secs
[09/24/26 16:07:45] INFO Created 770 records raw.py:723

=== Grounded answer ===
Based on the burn severity analysis of the Terzigno wildfire (August 7-18, 2025) using Sentinel-2 imagery within the specified bounding box (14.425,40.792 to 14.510,40.850), here are the key findings:

## Total Burned Area
**762.6 hectares** burned in total during the wildfire event.

## Burn Severity Breakdown
- **Low severity**: 546 patches covering 409.33 ha (53.7% of burned area)
- **Moderate-low severity**: 137 patches covering 285.18 ha (37.4% of burned area)
- **Moderate-high severity**: 87 patches covering 68.09 ha (8.9% of burned area)

## Priority Areas for Erosion Control
For erosion-control work before the rainy season, priority should be given to **moderate-high severity burned areas**, which pose the highest erosion risk due to:
- Significant vegetation loss and ground cover reduction
- Exposed mineral soil with low infiltration capacity
- Increased surface runoff and sediment yield potential

The analysis identified **25 priority patches** totaling **54.21 hectares** that meet the criteria for immediate erosion control intervention. These represent the most critical areas within the moderate-high severity zones (likely filtered by minimum patch size >0.5 ha and potentially considering factors like slope, proximity to waterways, or infrastructure).

## Recommendations
1. **Immediate focus**: Target the 54.21 ha of priority patches for erosion control measures (e.g., straw wattles, contour felled logs, hydroseeding, or check dams) before the first significant rainfall.
2. **Secondary focus**: The remaining 13.88 ha of moderate-high severity patches should be addressed as resources allow.
3. **Monitoring**: Establish sediment monitoring points in drainages below priority areas to evaluate treatment effectiveness.
4. **Long-term**: Consider reforestation with native species in moderate-low and low severity areas to restore vegetative cover and reduce future erosion risk.

The moderate-high severity zones (particularly the identified priority patches) represent the areas where erosion control investments will yield the greatest reduction in post-fire sedimentation and watershed degradation before the rainy season begins.
(base) alirezahamoudzadeh@Alirezas-MacBook-Air Demo %
```